

POLITECNICO DI TORINO

Master's Degree in Electronics



Master's Degree Thesis

Loss analysis of a 4-level, flying capacitor, bridge-less, totem-pole PFC

Supervisors

Prof. Francesco MUSOLINO

Ing. Marco PALMA

Candidate

Giorgio ARMIGNACCO

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Chapter 1

ZVS/ZCS modulations

Switching losses are by far the most relevant contributor to overall device efficiency, Slow-Leg MOSFET already implements a ZVS/ZCS technique to avoid switching the device on or off while having both current or voltages greater than 0.

The main inductor current modulation is done in such a way that near the 0 voltage cross of the 50Hz cycle, before the device turn-off, no current is flowing in the whole circuit (in order to maximize the power factor), also a small amount of current is fed through the device before the turn-on of the MOS channel, in order to pre-discharge the device output capacitances.

The same can not be said by the Fast-Leg devices, those ones are operating in an hard switching manner, causing a great amount of losses, which leads to the device heat-up and consequently reduction of longevity and reliability, at the same time they are limiting the system performances. A modulation technique called iTCM is proposed in order to achieve ZVS.

1.1 iTCM

1.1.1 TCM principle of operation

The Triangular Current Mode, TCM for short, is a type of modulation which forcibly induces Boundary Conduction Mode, BCM for short.

The inductor current actually reaches slightly negative values in order to provide enough charge to the FETs output capacitance, in order to discharge them before the channel turn-on, the peak current needed to achieve this is given by $I_{ZVS} = \frac{Q_o}{t_d}$ where t_d represents the amount of time available before the device turn-on, which by design will be the dead-time between the upper and lower FET, $Q_o = C_o V_{ds}$ where C_o is the FET output capacitance and V_{ds} is the voltage which the device is subject to, doing so a ZVS is achieved.

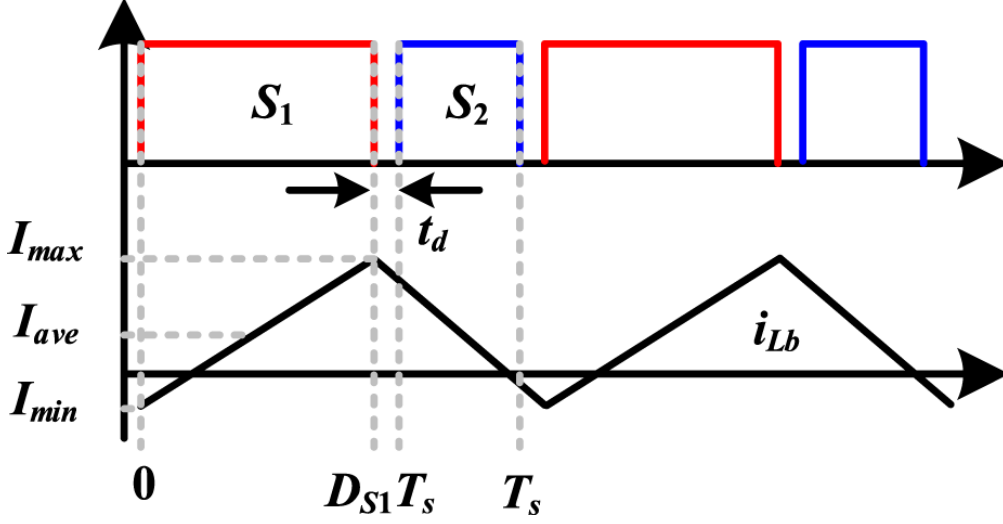


Figure 1.1: TCM current

This modulation scheme is a variation of the Pulse Width Modulation (PWM), the duty cycle, defined as $D = \frac{t_{on}}{T_s}$, is directly correlated with the system conversion ratio $M = \frac{V_o}{V_{in}}$ which is correlated to the conduction angle of the 50Hz cycle (V_{in} variation) and the system power output (V_o variations due to output capacitance faster discharge). The PWM modulation is responsible for the hard switching behaviour though, since it does not take into account the status of the FETs during a commutation.

The TCM modulation achieves the ZVS not by changing the duty cycle, but by changing the switching frequency $f_s = \frac{1}{T_s}$, doing so also the T_{on} time increases, giving more or less conduction time to the switches in order to reach a near 0 current in the circuit.

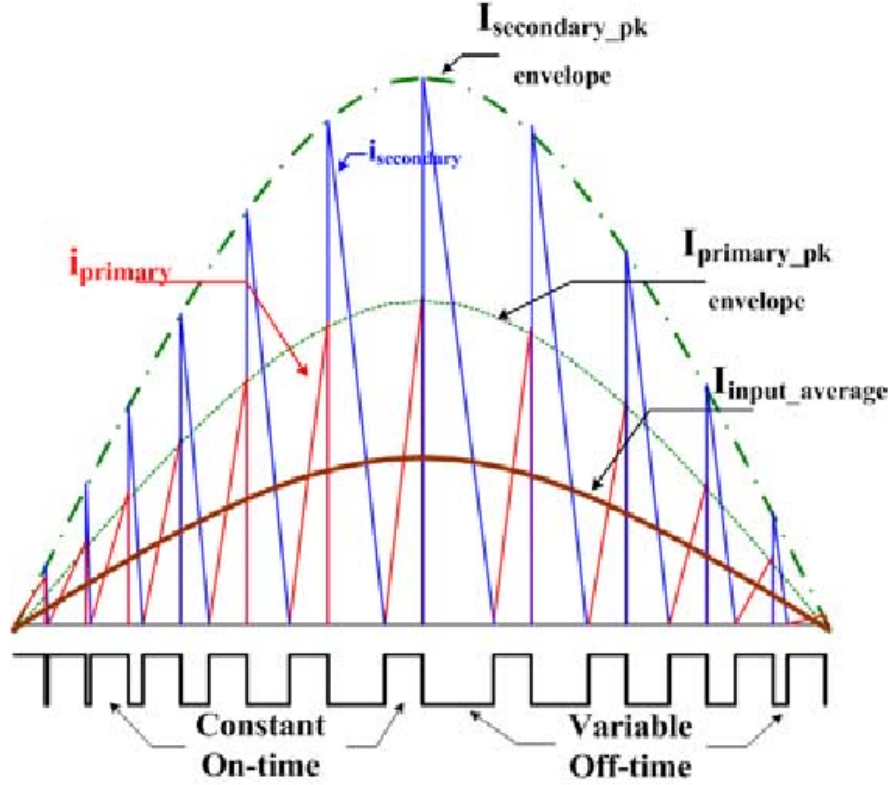


Figure 1.2: TCM frequency variation

1.1.2 TCM drawbacks

Inducing a TCM on the inductor current has its own drawbacks on the system performances, at first there is the inductor design itself which becomes more demanding since a higher current ripple is imposed on the inductor, which leads to higher AC losses, caused by the core magnetization, which consequently causes higher heat output and overall loss of efficiency.

Switches and capacitors also experience the same current ripple, and need to withstand a higher instantaneous peak current to achieve the same system power output.

This leads to an overall increase of the system stress.

Another important aspect is the worsening of the $iTHD$, and consequently the Power Factor, since this higher current ripple must go through the AC line, which must provide the reactive power for the inductor to be charged/discharged.

Both arguments are addressed by the iTCM modulation technique.

1.1.3 iTCM

The integrated Triangular Current Mode, iTCM, is very similar to the TCM modulation scheme but an additional LC branch is added to decouple the role of the inductor providing energy to the output, from the one that must provide energy to the FETs in order to discharge them.

The reasoning behind the iTCM modulation is simple, keep a CCM on the main inductor, while providing BCM for the switches.

Starting from the totem-pole-bridgeless topology, we have a main inductor L_g which is responsible to provide energy to the whole system, currently this inductor is NOT reaching a 0 current condition UNLESS we are near the 0 crossing condition of the 50Hz cycle (Power Factor Correction).

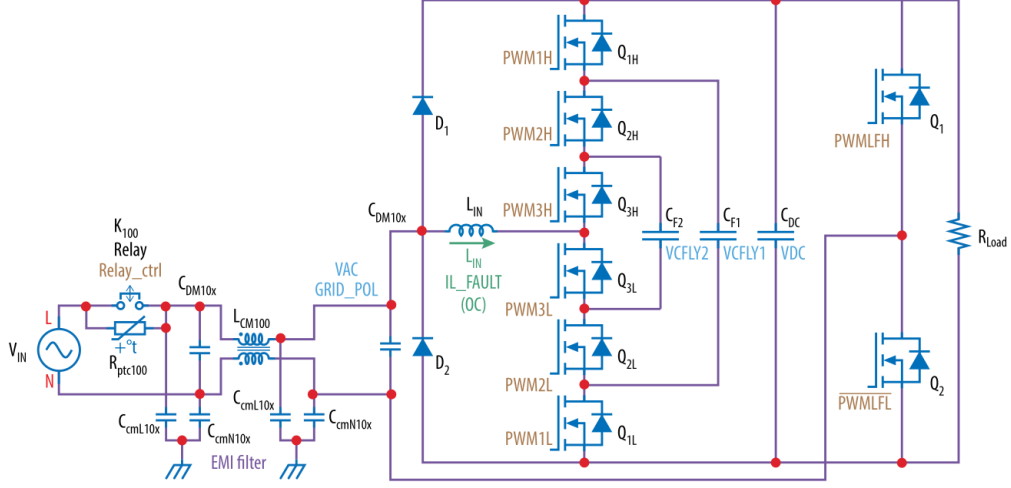


Figure 1.3: PFC with CCM behaviour

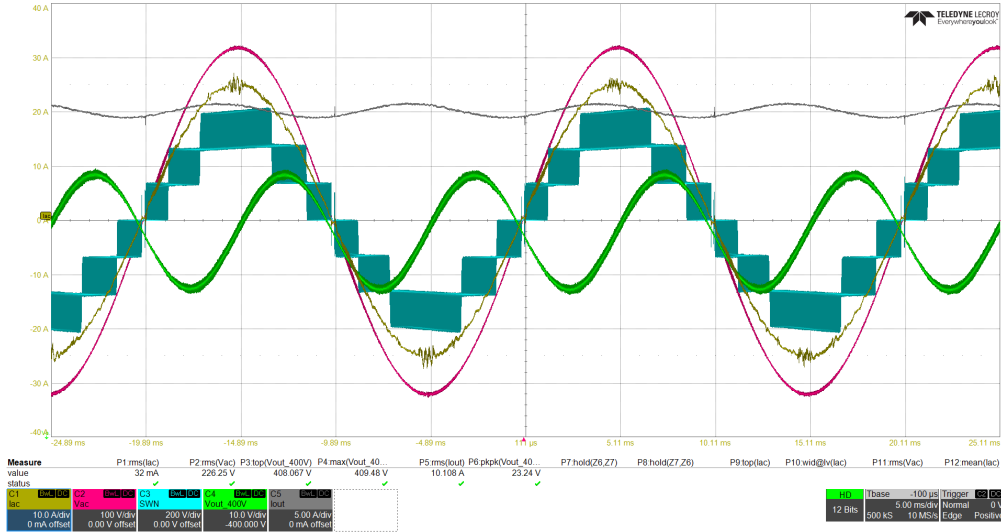
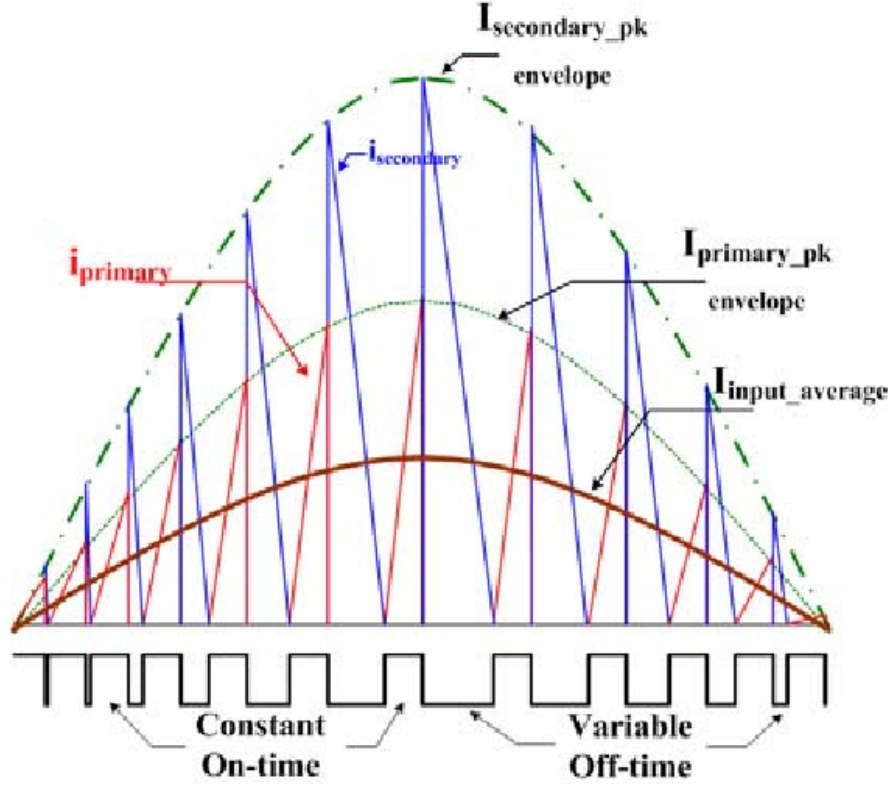


Figure 1.4: L_g current behaviour while in CCM

On the L_g inductor there is currently a superimposition of the 50Hz Low Frequency (LF) current trend, plus the High Frequency (HF) f_s current ripple of the converter. Inducing a TCM in the main inductor will cause the average input current to stay the same (only depends on system power demands) but the current peaks are much higher.


 Figure 1.5: L_g current behaviour while in BCM

Let's build the iTCM circuit step-by-step

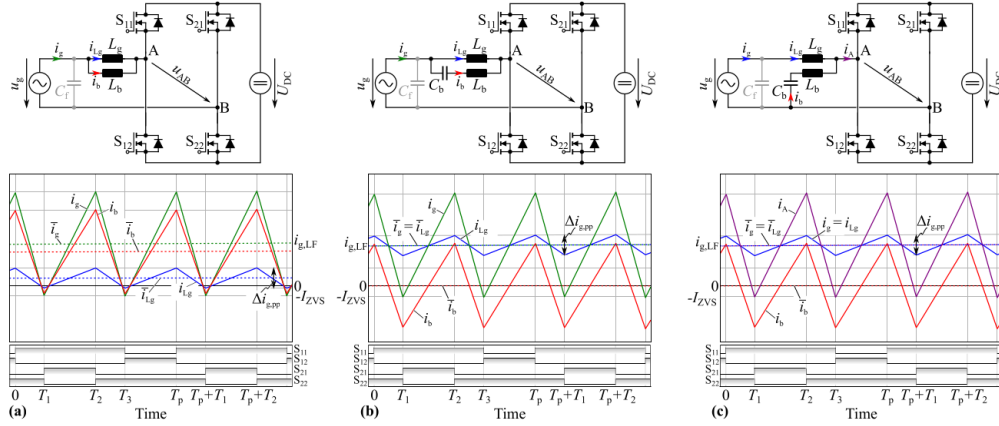


Figure 1.6: iTCM circuit build up

By adding an inductor L_b much smaller than the main inductor L_g the overall inductance is governed by L_b , causing the current ripple to drastically increase. The $L_b + L_g$ inductor is now responsible for both the system output current and the ZVS current, indistinguishable from the

TCM modulation.

We add a capacitor C_b in series to the L_b inductor, this creates an high-pass filter, now the L_b inductor will not be responsible for the 50Hz cycle current trend, this will also remove any kind of DC bias on the current flowing through L_b , if designed correctly this high-pass filter will only let the f_s current ripple of the converter through while leaving all lower frequency components out, which will be carried by the L_g inductor since they have no other path.

The L_g inductor still experiences a small HF current ripple, but it is working in CCM, so the core stresses are greatly reduced since it experiences a much smaller ripple, most of the current ripple is managed by the L_b inductor, since the current finds a lower impedance path through it.

Doing so we effectively decoupled the role of the 2 inductors, the main one L_g is the one responsible to provide power to the system, and has to be designed to achieve an high flux density and filling factor, while the auxiliary one L_b is responsible for the ZVS behavior, and can be optimized for low flux high frequency behaviour.

The iTCM controller will monitor the system status in order to make sure that the L_b inductor is reaching $-I_{ZVS}$ current at all transitions in order to discharge the FET capacitances, by modulating the switching frequency.

The I_{ZVS} current is still forced to flow through the AC line, which in the end will negatively impact the iTHD and the consequently the PF.

It can be noted that both the L_g and $L_b + C_b$ branch experience the same voltage swings, by placing this LC branch in parallel to the AC input, we effectively decouple the HF components from the LF ones, this is the meaning of the i of "integrated TCM" because all the HF current ripple is kept inside the converter.

Doing so the iTHD performance are not negatively affected by the TCM modulation.

The C_b capacitor will experience a voltage which will be the same as the AC voltage line, but its capacitance is rather small, since the frequency of the LC branch is pretty high.

1.1.4 iTCM inductor design guidelines

Let's consider the following circuit, which represents an iTCM inverter, which is a DC/AC converter, while a PFC is a AC/DC converter. The hardware layout is exactly the same of a totem-pole bridge-less PFC and his behavior can be studied in order to understand the math behind the iTCM.

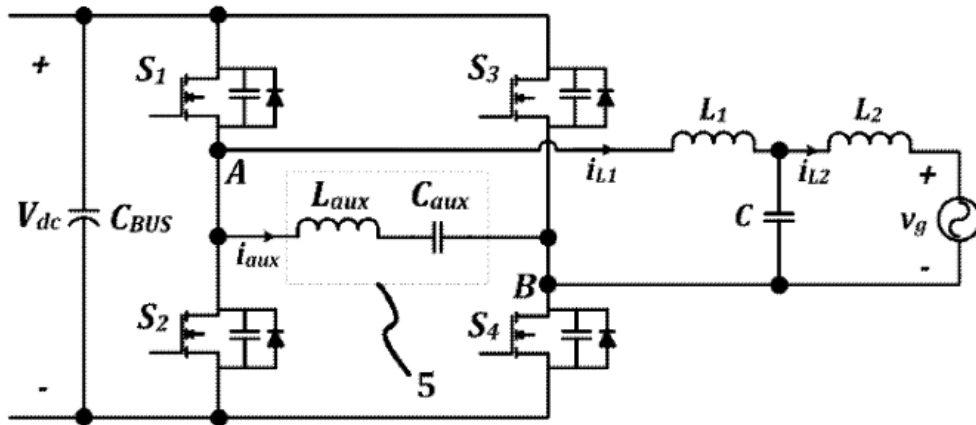


Figure 1.7: iTCM inverter

In this schematic $L_1 + L_2$ represents L_g , while L_{aux} represents L_b . It must be noted that L_{aux} does not have any DC bias, while L_g is following the 50Hz trend in order to provide energy to the output, but in the following figures I_{L_g} is assumed constant for the sake of discussion, since we are considering a very narrow portion of the 50Hz cycle.

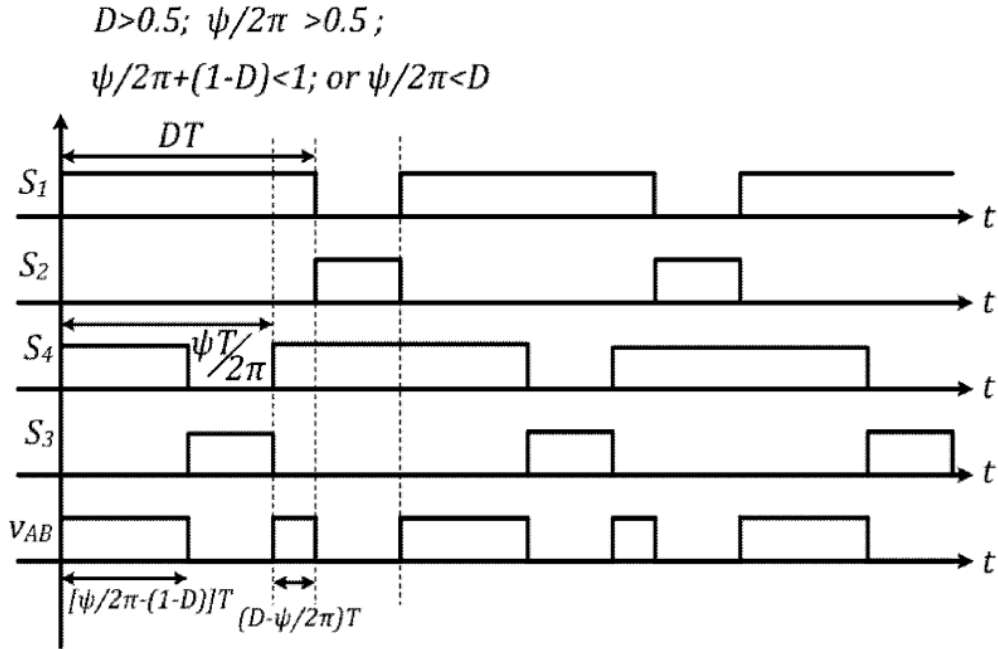


Figure 1.8: iTCM inverter switching scheme

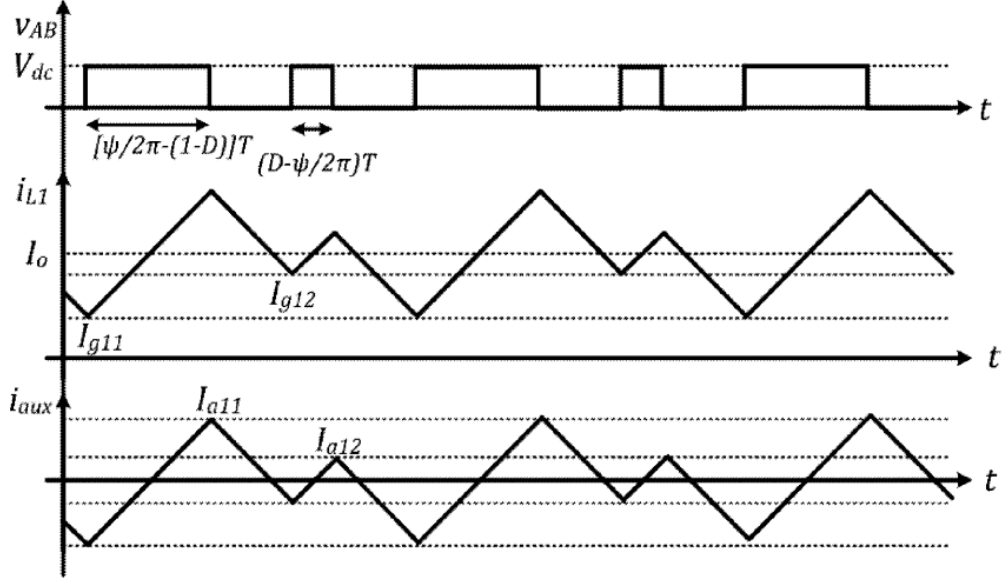


Figure 1.9: iTCM inverter inductor currents

4 quantities must be noted, I_{g11} , I_{g12} , I_{a11} , I_{a12}

I_{g11} is the amount of current flowing in the main circuit, which is the one providing energy to the output, in order to achieve a ZVS the auxiliary inductor must reach an amount of current which is $I_{a11} = I_{g11} + I_{ZVS}$.

Analyzing the circuit topology we get

$$\begin{aligned}
 I_{a12} &= \frac{V_{dc}}{L_{aux}f}(1-D)\left[D - \frac{\Phi}{2\pi}\right] \\
 I_{g12} &= \left[I_o - \frac{V_{dc} - V_g}{2L_gf}\left(D - \frac{\Phi}{2\pi}\right)\right] \\
 I_{a11} &= \frac{V_{dc}}{L_{aux}f}(1-D)\left[\frac{\Phi}{2\pi} - (1-D)\right] \\
 I_{g11} &= I_o - \frac{V_{dc} - V_g}{2L_gf}\left[\frac{\Phi}{2\pi} - (1-D)\right] \\
 I_{ZVS} &= \frac{V_{DS}C_{DS}}{t_d}
 \end{aligned} \tag{1.1}$$

where f is HF ripple frequency of the inductor and not the switching frequency, Φ is phase shift between the 2 active regions of the inverter V_{dc} is the DC bus link and V_g is the output voltage, which is assumed to be constant at this instant of time for the sake of discussion, L_{aux} is the auxiliary inductance, D is the duty cycle

In order to achieve BCM (including I_{ZVS}) on the L_{aux} the optimal condition is reached for $I_{a11} = I_{a12}$ which is obtained for $\Phi = \pi$, analyzing the MOS currents it's clear that the MOS currents goes below 0.

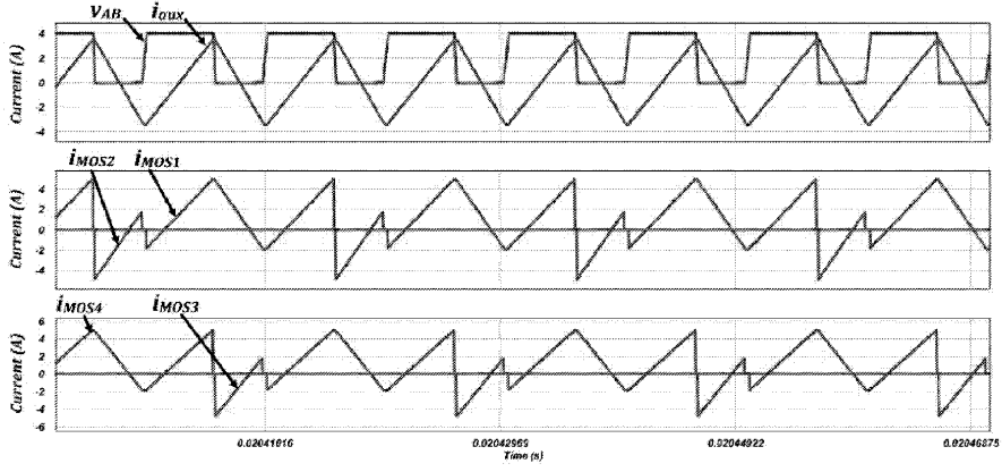


Figure 1.10: iTCM inverter MOS currents

Let's flip the script and consider a totem-pole bridge-less PFC

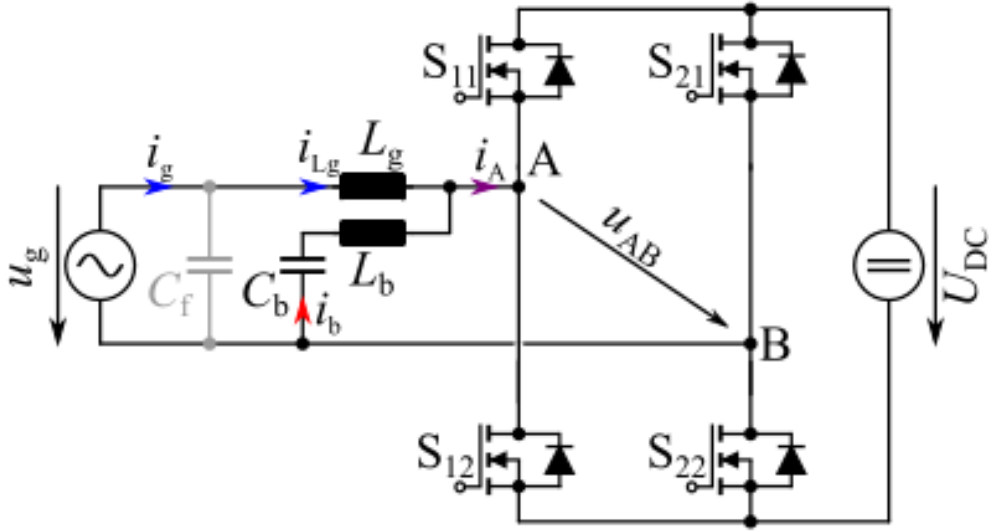


Figure 1.11: iTCM PFC

we can calculate the ENVELOPES of I_g and I_b

$$\begin{aligned}
 V_{pk} &= V_{AC_{RMS}} \sqrt{2} \\
 V_g(t) &= V_{pk} |\sin(\omega t)| \left[1 - \frac{V_{pk}}{V_{DC}} |\sin(\omega t)| \right] \\
 A(t) &:= \frac{V_g(t)}{V_{pk}}
 \end{aligned} \tag{1.2}$$

$$\begin{aligned}
 V_{pk} &= V_{AC_{RMS}} \sqrt{2} \\
 V_g(t) &= V_{pk} |\sin(\omega t)| \left[1 - \frac{V_{pk}}{V_{DC}} |\sin(\omega t)| \right]
 \end{aligned}$$

$$A(t) := \frac{V_g(t)}{V_{pk}}$$

$$I_{g_{env}} = \frac{2P_o}{V_{pk}} + \frac{1}{2} \frac{V_{pk} A(t)}{L_g} \frac{1}{f_{eff}} \quad (1.3)$$

$$I_{b_{env}} = \frac{1}{2} \frac{1}{f_{eff}} \frac{V_{pk}}{L_b} \quad (1.4)$$

in order to achieve ZVS the current trough the auxiliary inductor L_b must "close the gap" between the minimum of the current flowing through L_g and $-I_{zvs}$

imposing the condition

$$|max(I_{b_{env}}) - min(I_{g_{env}})| \equiv I_{ZVS}$$

we can manipulate those equations in order to calculate the values for L_g and L_b , by inverting (1.3) and (1.4) we get

$$\begin{aligned} L_g &= \frac{V_{pk}^2}{2r f_{min} P_o} \left(1 - \frac{V_{pk}}{V_{DC}}\right) \\ I_{ZVS} &= \frac{V_{DS} C_{DS}}{t_d} \\ L_b &= \frac{V_{pk}^2}{2(2-r) f_{min} P_o + 2I_{ZVS} V_{pk}} \left(1 - \frac{V_{pk}}{V_{DC}}\right) \end{aligned} \quad (1.5)$$

were $r = \frac{\Delta I_{gPP}}{I_{gpk}}$ this will allow us to define all the major components.

For the capacitor C_b we simply define that the $Z|_{f_{min}} = \frac{Z_b}{d}$ where Z_b is the natural frequency of the LC resonator

$$\begin{aligned} I_{B_{pk}} &= \frac{V_{pk}}{2f_{min} L_b} \left(1 - \frac{V_{pk}}{V_{DC}}\right) \\ C_b &\geq \frac{I_{B_{pk}}}{4\pi^2 f_{min}^2 L_b I_{ZVS} d} \end{aligned} \quad (1.6)$$

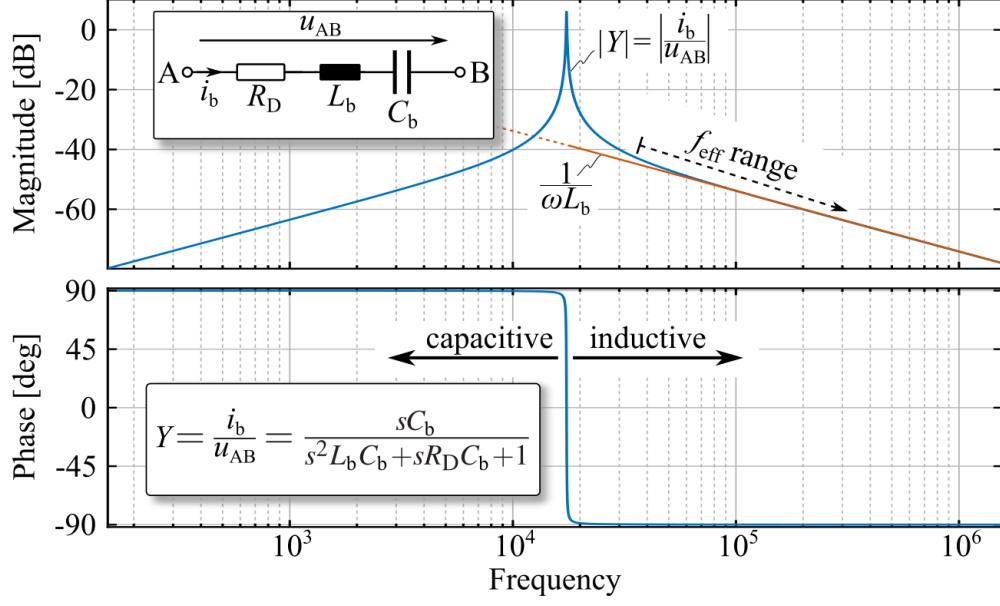


Figure 1.12: C_b frequency behaviour

Another important aspect of the TCM and by consequence the iTCM modulation is the fact that the frequency of the modulation can be modified, by doing so we can decouple the role of the duty cycle from the role of the frequency modulation, the duty cycle will be mainly imposed by the load, while the frequency is imposed by the angle of the AC grid.

Doing so the amount of time from which the devices are in conduction is now defined by 2 commands, which are governed by 2 separate variables of the system.

Modifying the frequency has also another advantage, the current ripple on the L_b inductor is minimized, so also the stress on that component is reduced

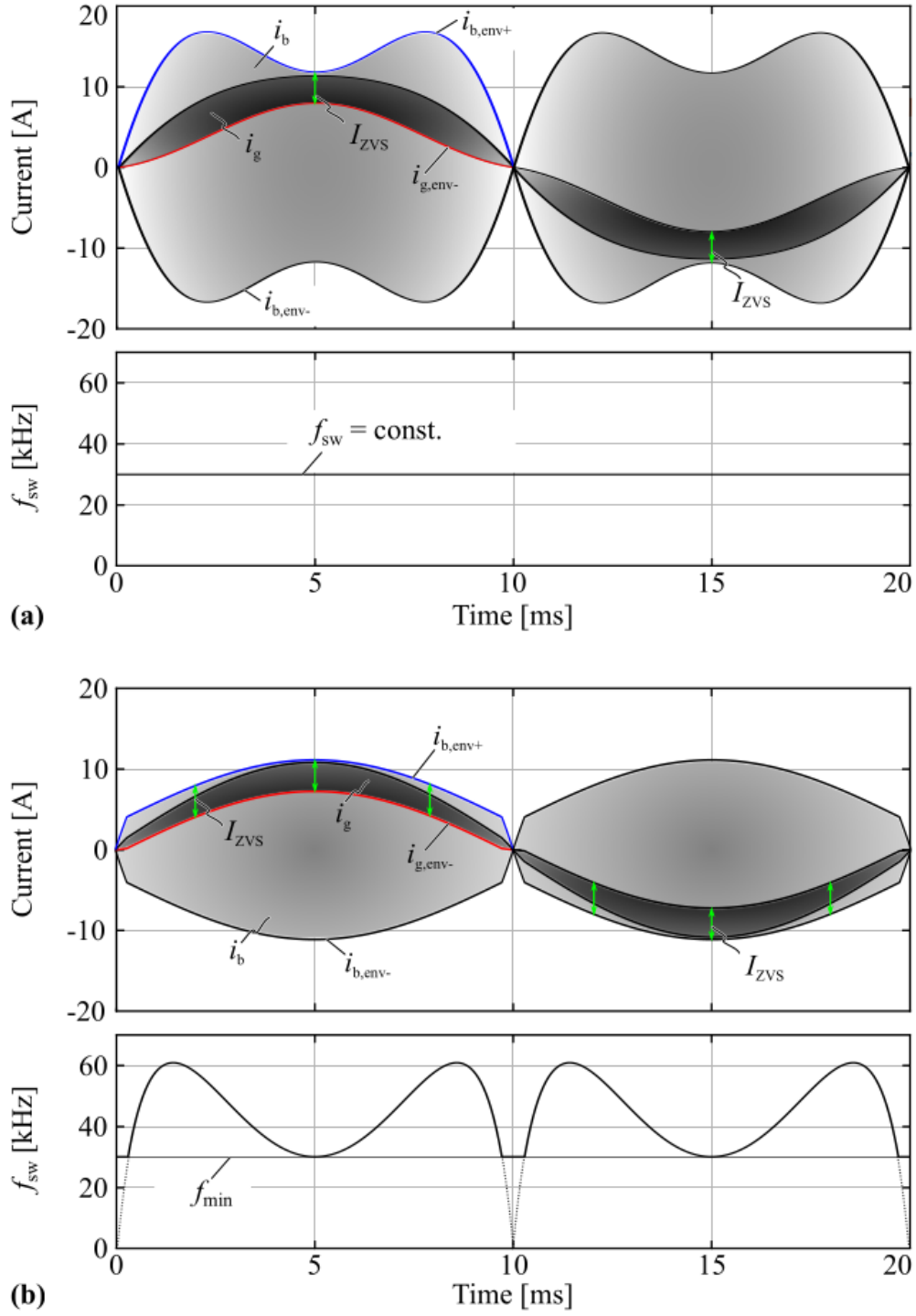


Figure 1.13: iTCM frequency modulation effect

The frequency of the inductor can be calculated as follow , recalling 1.2

$$f = \frac{A(t)V_{pk}}{4P_o|\sin(\omega t)| + 2V_{pk}I_{ZVS}}\left(\frac{1}{L_b} + \frac{1}{L_g}\right) \quad (1.7)$$

by calculating the derivative we get that we have a maxima for $\omega t = \frac{12}{5}$ and a minima for $\omega t = 1$, replacing those values we can get the maximum and minum frequency at which the inductor is subject to.

1.2 Design criteria for EPC91107

For the EPC91107 we have the following criteria

$$\begin{aligned} V_{AC} &= 230V_{RMS} \\ V_{DC} &= 400V \\ N &= 4 \\ P_o &= 5kW \\ r &= 40\% \\ f_{min} &= 360.7kHz \end{aligned} \quad (1.8)$$

the value of 40% has been documented in all previous papers to be proven that is a great compromise between ripple current and inductor size.

The value of L_g r and f_{min} are actually tied together, the current value of r computed with the elements mounted on the circuit is 34.35%, by relaxing a bit this constrain we can also decrease the f_{min} value, this will allow us to relax a bit the timing of the GaNFETs, with this reasoning we can define

$L_g = 13.7\mu H$ moving on with I_{ZVS}

$$\begin{aligned} V_{DS} &= \frac{V_{DC}}{(N-1)} = 135V \\ C_{DS} &= 1nF(EPC2304) \\ t_d &= 17ns \\ I_{ZVS} &= \frac{V_{DS}C_{DS}}{t_d} = 7.94A \end{aligned} \quad (1.9)$$

$$\begin{aligned} L_b &= \frac{V_{pk}^2}{2(2-r)f_{min}P_o + 2I_{ZVS}V_{pk}}\left(1 - \frac{V_{pk}}{V_{DC}}\right) = 2.59\mu H \\ L_b &= 2.7\mu H \end{aligned} \quad (1.10)$$

by replacing the values to find minima and maxima of 1.7 we get

$$\begin{aligned} f_{min} &= 360.7kHz \\ f_{max} &= 764.742kHz \\ f_{sw_{min}} &= \frac{f_{min}}{(N-1)} = 120.23kHz \\ f_{sw_{max}} &= \frac{f_{max}}{(N-1)} = 254.914kHz \end{aligned} \quad (1.11)$$

From 1.6 imposing $d = 10\%$ we get

$$\begin{aligned} I_{B_{pk}} &= 31.97A \\ C_b &\geq 2.83\mu F = 3.3\mu F @ 400V \end{aligned} \tag{1.12}$$

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